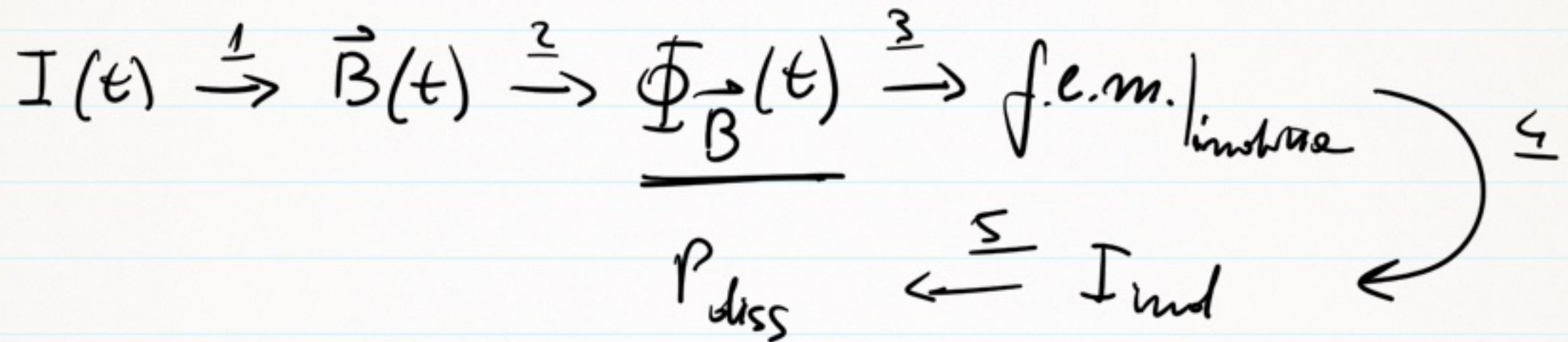
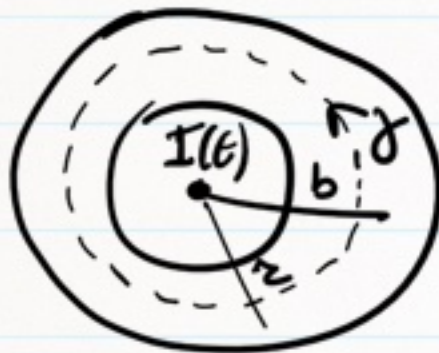


09/12/20

Ex. 1



1 trovare B : legge di Ampère (legge di Biot-Savart)



$$\oint_{\gamma} \vec{B} \cdot d\vec{\ell} = \mu_0 I$$

$$B \cdot 2\pi r = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}, \quad \vec{B}(r) = \frac{\mu_0 I}{2\pi r} \hat{u}_{\varphi}$$

2 trovare $\Phi(\vec{B})$: flusso concatenato col solenoide

$$\Phi(\vec{B}) = \iint_S \vec{B} \cdot \vec{n} \, dS \stackrel{1}{=} \iint_S B(r) \vec{u}_\varphi \cdot \vec{u}_\varphi \, dS$$

$(\vec{n} \equiv \vec{u}_\varphi)$

$$\simeq B(r=b) \cdot S \cdot N$$

$$\Phi(\vec{B}) = \frac{\mu_0 I S N}{2\pi b}$$

3 trovare $\mathcal{E}_i$: Legge di Faraday

$$\mathcal{E}_i = - \frac{d\Phi(\vec{B})}{dt} = - \frac{d}{dt} \left(\frac{\mu_0 I(t) SN}{2\pi b} \right) =$$

$$= - \frac{\mu_0 SN}{2\pi b} K, \quad \text{per } t > 0$$

4 trovare I_i : Legge di Ohm

$$I_i = \frac{\mathcal{E}_i}{R} = - \frac{\mu_0 SN}{2\pi b R} K$$

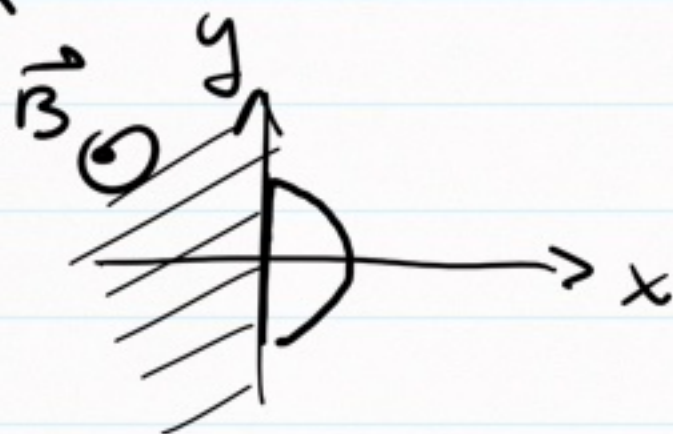
5 trovare P_{diss} : $P_{diss} = \mathcal{E}_i \cdot I_i = I_i^2 \cdot R =$

$$= \left(\frac{\mu_0 SN K}{2\pi b} \right)^2 \cdot \frac{1}{R}$$

Ex. 2

(a) $I_i = \frac{\Sigma_i}{R}$, $\Sigma_i = - \frac{d\Phi(\vec{B})}{dt} \leftarrow \underline{S = S(t)}$

@ $t = 0$



per $t > 0$?

$\theta = \omega t$

$\Phi(\vec{B})$

increases $0 < \theta < \pi$

decreases $\pi < \theta < 2\pi$

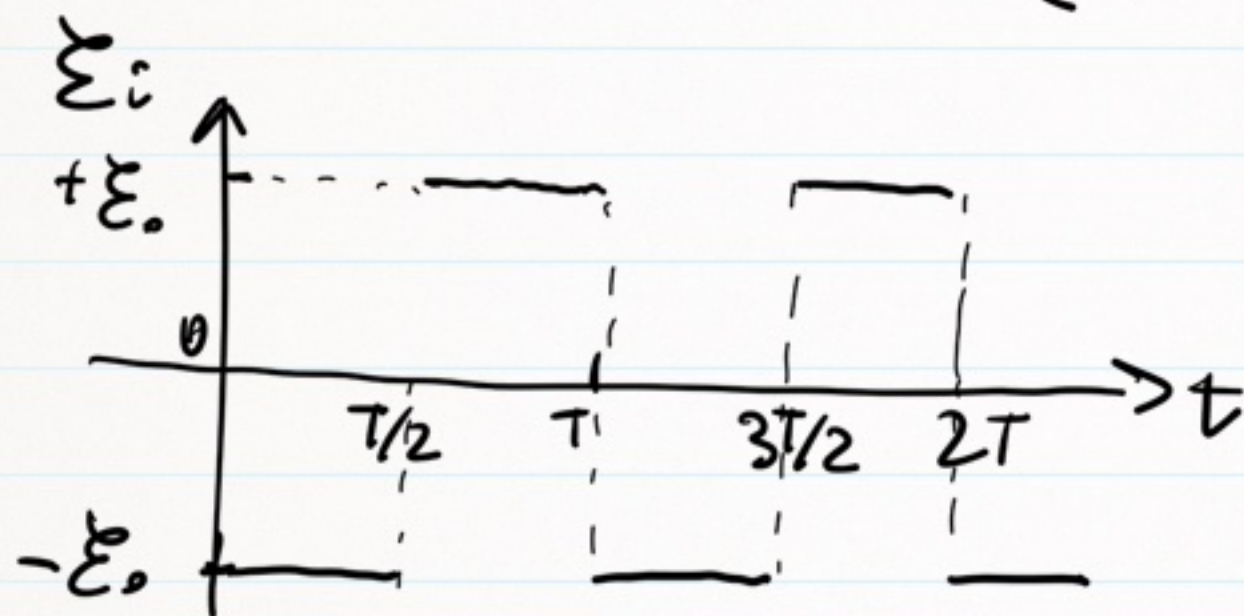
e periodicamente
si ripete

$$S(t) = \begin{cases} \frac{\pi a^2}{2} \cdot \frac{\omega t}{\pi} & , \quad 2n\pi < \omega t < (2n+1)\pi, \quad \forall n \in \mathbb{N} \\ \frac{a^2}{2} (-\omega t - 2\pi) & , \quad (2n+1)\pi < \omega t < (2n+2)\pi, \quad \forall n \in \mathbb{N} \end{cases}$$



$$\Phi_B(t) = B \cdot S(t)$$

$$\Sigma_i = - \frac{d\Phi_B(t)}{dt} = \begin{cases} -\frac{B a^2 \omega}{2}, & 2n\pi < \omega t < (2n+1)\pi \\ +\frac{B a^2 \omega}{2}, & (2n+1)\pi < \omega t < (2n+2)\pi \end{cases}$$

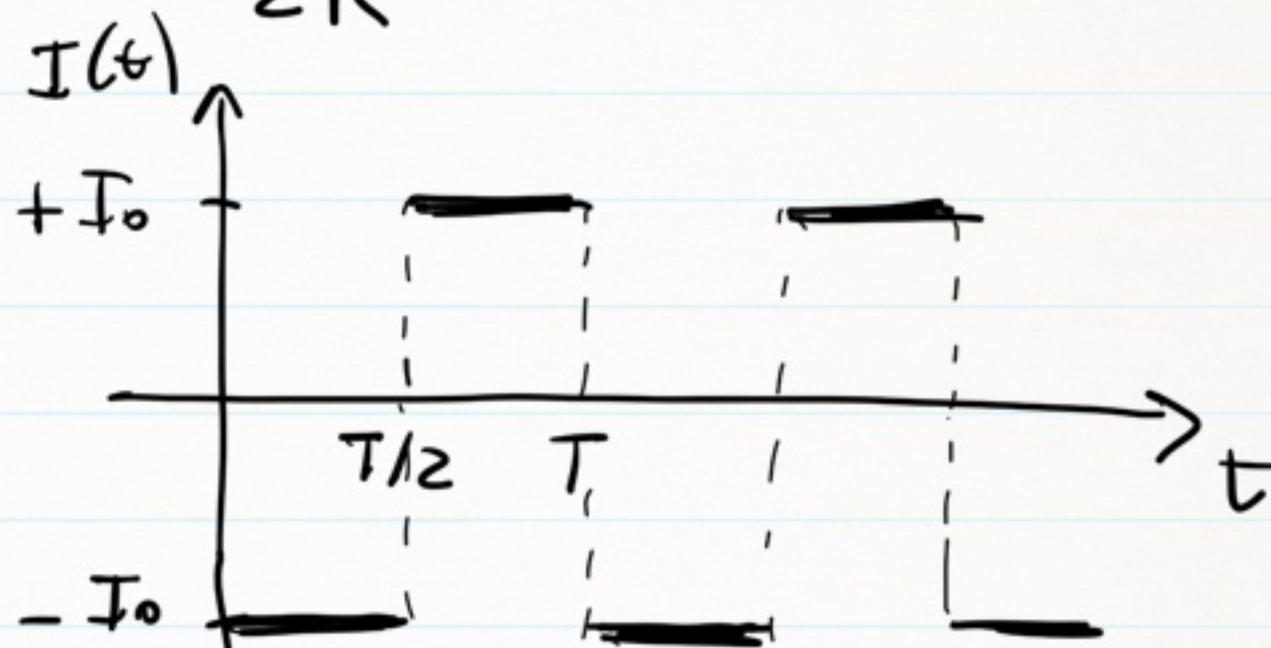


$$\left(\begin{aligned} \Sigma_0 &= B a^2 \omega / 2 \\ T &= \frac{2\pi}{\omega} \end{aligned} \right)$$

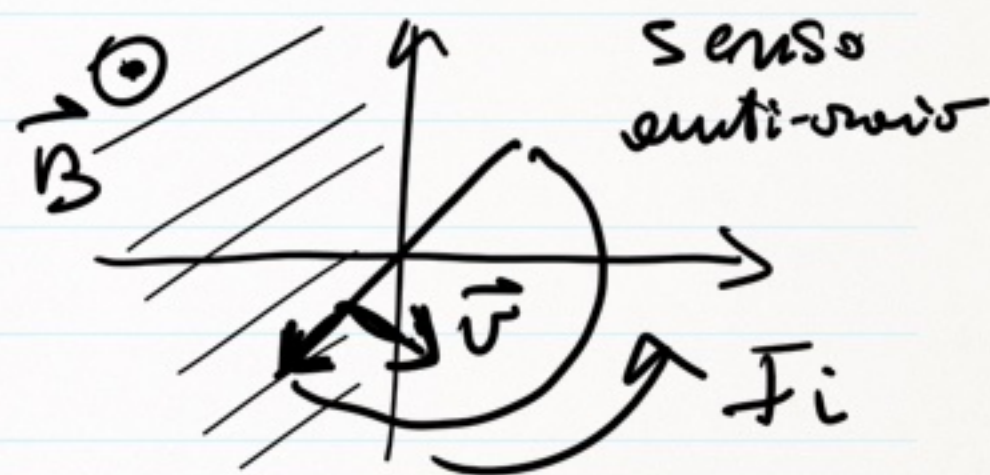
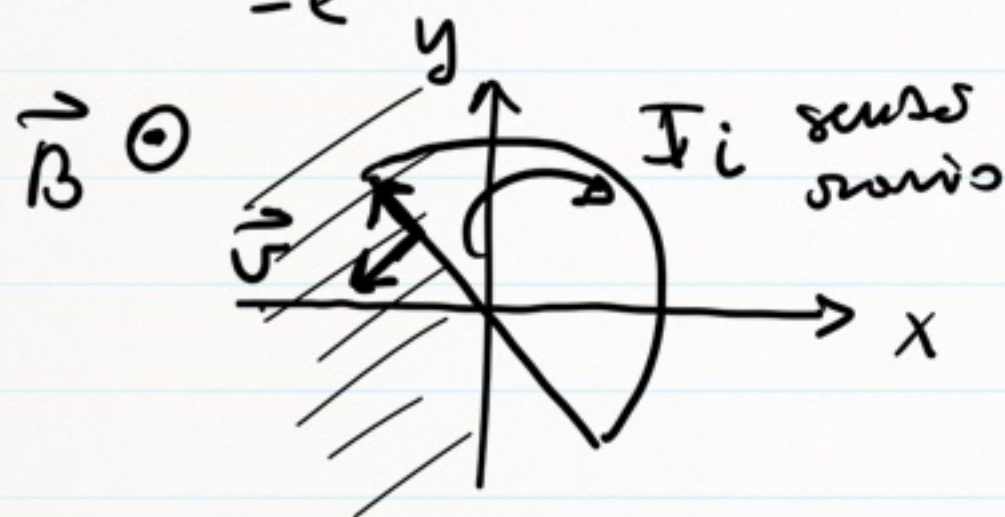
$$I_i(t) = \frac{\sum \varepsilon_i(t)}{R} = \begin{cases} -\frac{B_0^2 \omega}{2R} & 0 < t < T/2 \\ +\frac{B_0^2 \omega}{2R} & T/2 < t < T \end{cases}$$

e periodi successivi

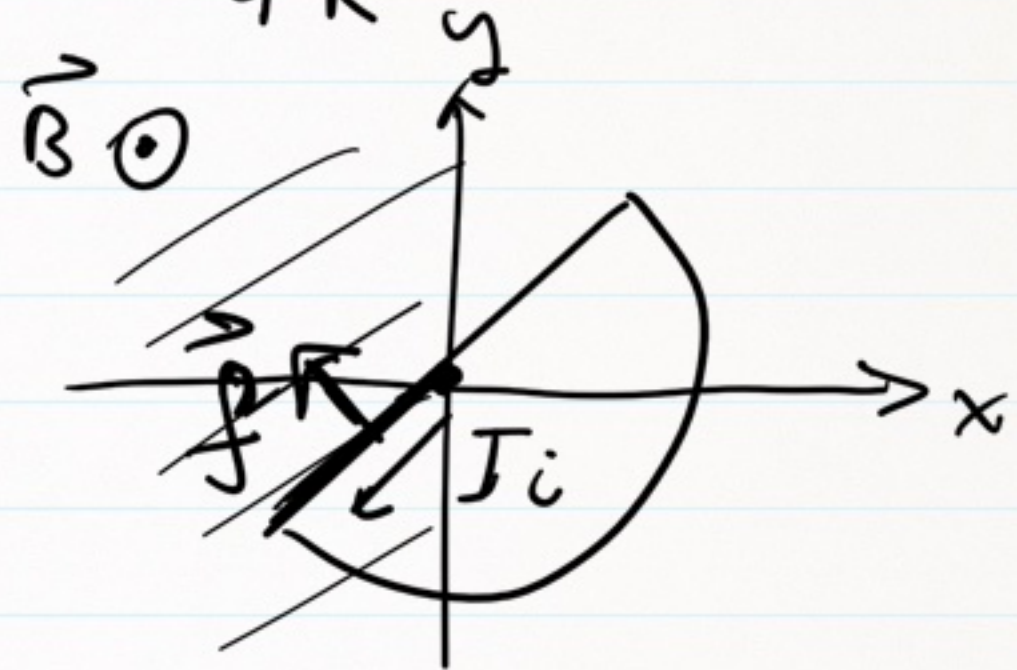
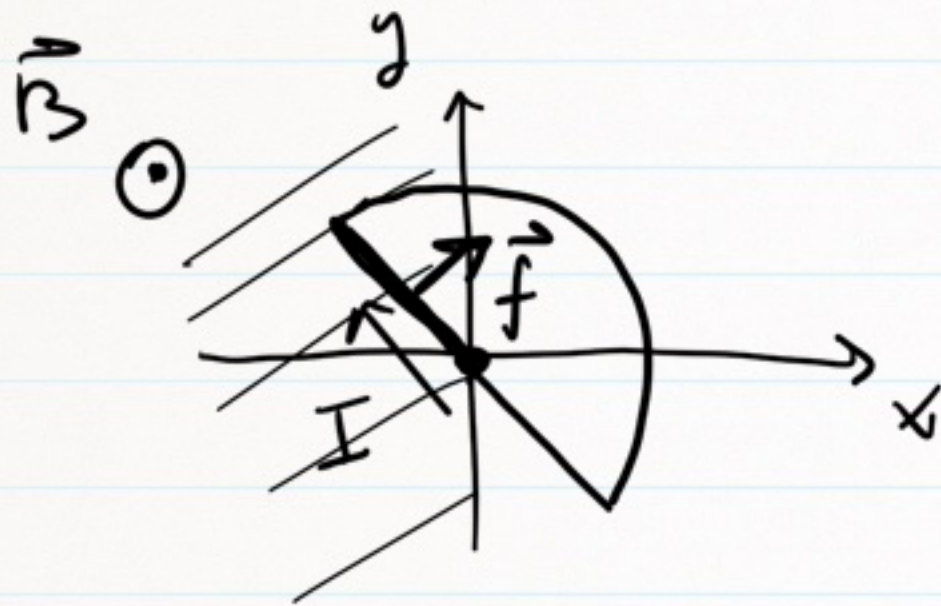
$$I_0 = B_0^2 \omega / 2R$$



$$\vec{E}_{ind} = \frac{F}{-e} = \vec{v} \times \vec{B}$$



$$(b) \quad \underline{P_{el, diss}} = R \cdot I_i^2 = \frac{B^2 a^4 \omega^2}{4R}$$



$$d\vec{f} = I_i d\vec{l} \times \vec{B}$$

$$d\vec{M} = \vec{r} \times d\vec{f}$$

$$\begin{aligned} \vec{M} &= \int_0^a \vec{r} \times d\vec{f} = -\vec{u}_z I_i \cdot B \int_0^a l dl = \\ &= -\frac{B^2 a^4 \omega}{4R} \vec{u}_z \end{aligned}$$

$$P_{mecc} = -\vec{M} \cdot \vec{\omega} = \frac{B^2 a^4 \omega^2}{4R} \equiv P_{el, diss} (!)$$

(c) senza autoinduzione: $\mathcal{E}_i = R \cdot I$

con autoinduzione: $\mathcal{E}_i + \mathcal{E}_L = R \cdot I$

$$\mathcal{E}_L = -L \cdot \frac{dI}{dt}$$

$$\left(\mathcal{E}_L = -\frac{d\Phi_L}{dt} = -\frac{d}{dt}(L \cdot I) \right)$$

$$\mathcal{E}_i - R I = L \cdot \frac{dI}{dt} \quad \Leftrightarrow \quad \text{equivalente} \quad \mathcal{E}_i \begin{array}{c} \text{---} R \text{---} \\ | \\ \text{---} L \text{---} \end{array}$$

- Risolvere per sep. variabili + condizione iniziale

$$\int \frac{dI}{\mathcal{E}_i - RI} = \int \frac{dt}{L}$$

$$\ln \mathcal{E}_i - RI = -\frac{R}{L}t + \cos t$$

$$\mathcal{E}_i - RI = e^{-\frac{R}{L}t + \cos t} = A \cdot e^{-t/\tau}$$

$$(\tau = L/R)$$

$$\underline{I(t)} = \frac{\mathcal{E}_i}{R} - \frac{\textcircled{A}}{R} e^{-t/\tau}$$

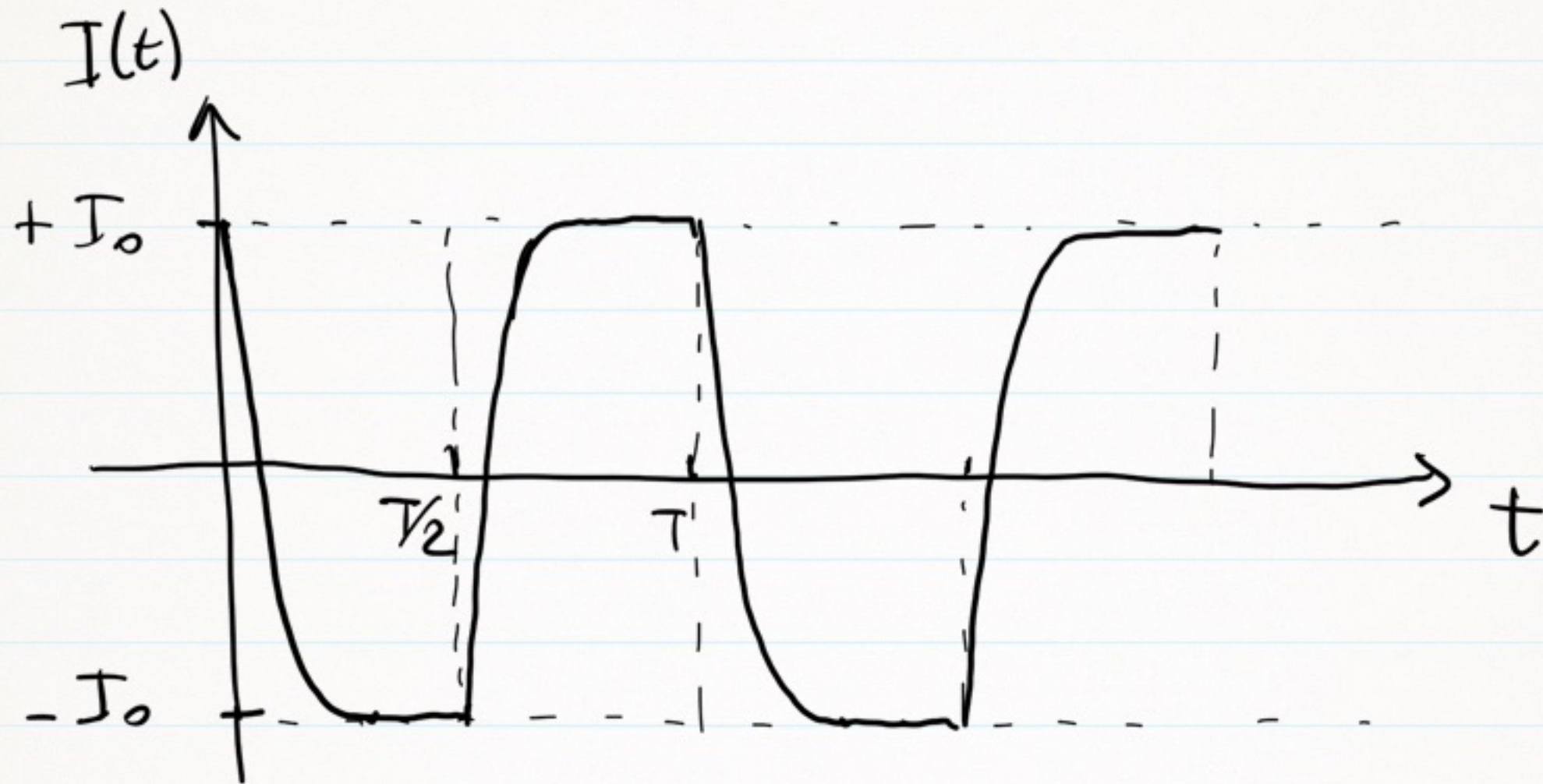
$$H_p: \quad \tau \ll T/\omega \quad \Rightarrow \quad I(t=\theta^+) \neq I(t=\theta^-)$$

$$\begin{cases} I(t=\theta^-) = I_0 = \frac{\mathcal{E}_0}{R} \\ I(t=\theta^+) = -\frac{\mathcal{E}_0}{R} - \frac{A}{R} \end{cases} \quad \rightarrow \quad -\frac{A}{R} = \frac{2\mathcal{E}_0}{R}$$

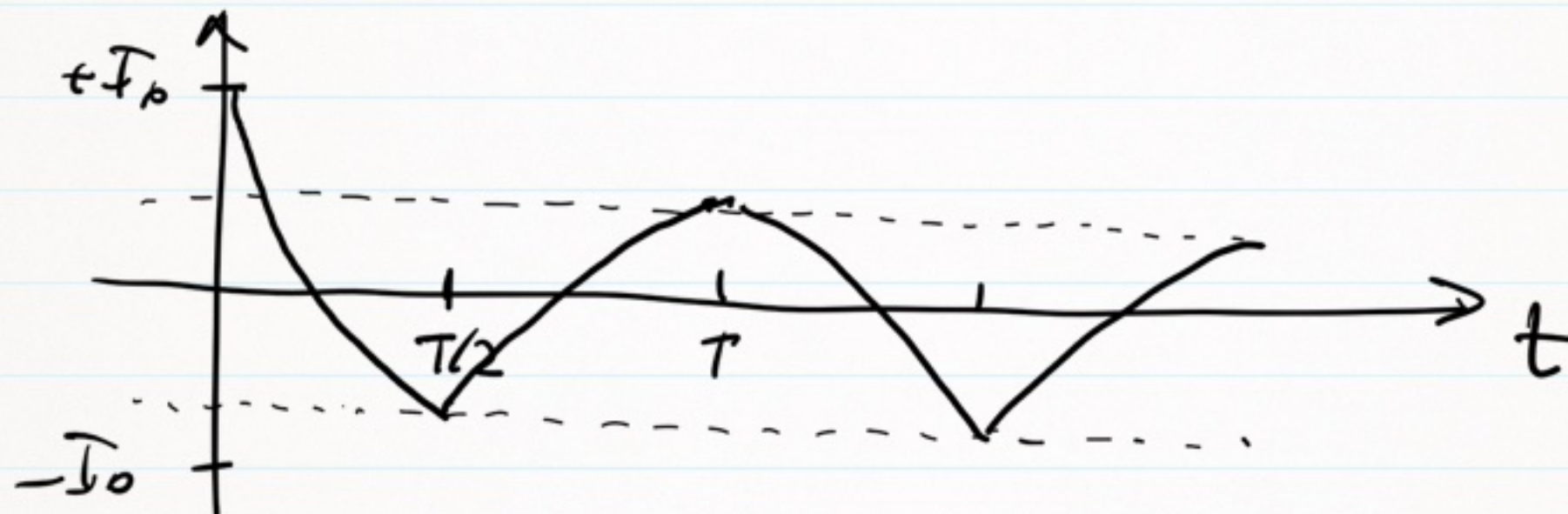
$$\text{per } 0 < t < T/2 : \quad I(t) = \frac{\mathcal{E}_0}{R} (2e^{-t/\tau} - 1)$$

$$\text{per } T/2 < t < T : \quad I(t) = -\frac{\mathcal{E}_0}{R} (2e^{-(t-T/\omega)/\tau} - 1)$$

 e multipli
 (periodicità)



se $H_p: \tilde{\omega} \ll \pi/\omega$ non fosse soddisfatta



Ex. 3

$$(a) \Phi(\vec{B}) = \Phi_B [z(t)] \underset{\substack{\text{campo} \\ \text{uniforme} \\ \text{sullo spire}}}{\simeq} \pi a^2 \cdot \frac{B_0}{L} (z_0 + vt)$$

$$\mathcal{E}_i = - \frac{d\Phi(\vec{B})}{dt} = - \frac{\pi a^2 B_0 v}{L}$$

$$I_i = - \frac{\pi a^2 B_0 v}{L R}$$

$$(b) P_{\text{Joule}} = R \cdot I_i^2 = \frac{\pi^2 a^4 B_0^2 v^2}{L^2 R}$$

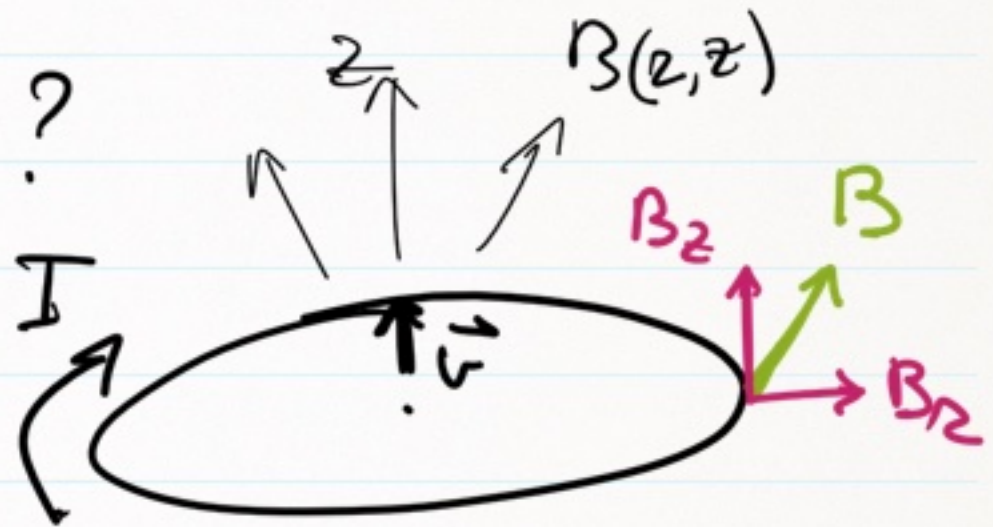
$$P_{mecc} = \vec{F}_{ext} \cdot \vec{v} = P_{joule} = \frac{(\pi e^2 B_0 v)^2}{L^2 R}$$

$$\vec{F} = - \vec{F}_{ext} = - \frac{(\pi e^2 B_0)^2}{L^2 R} \vec{v}$$

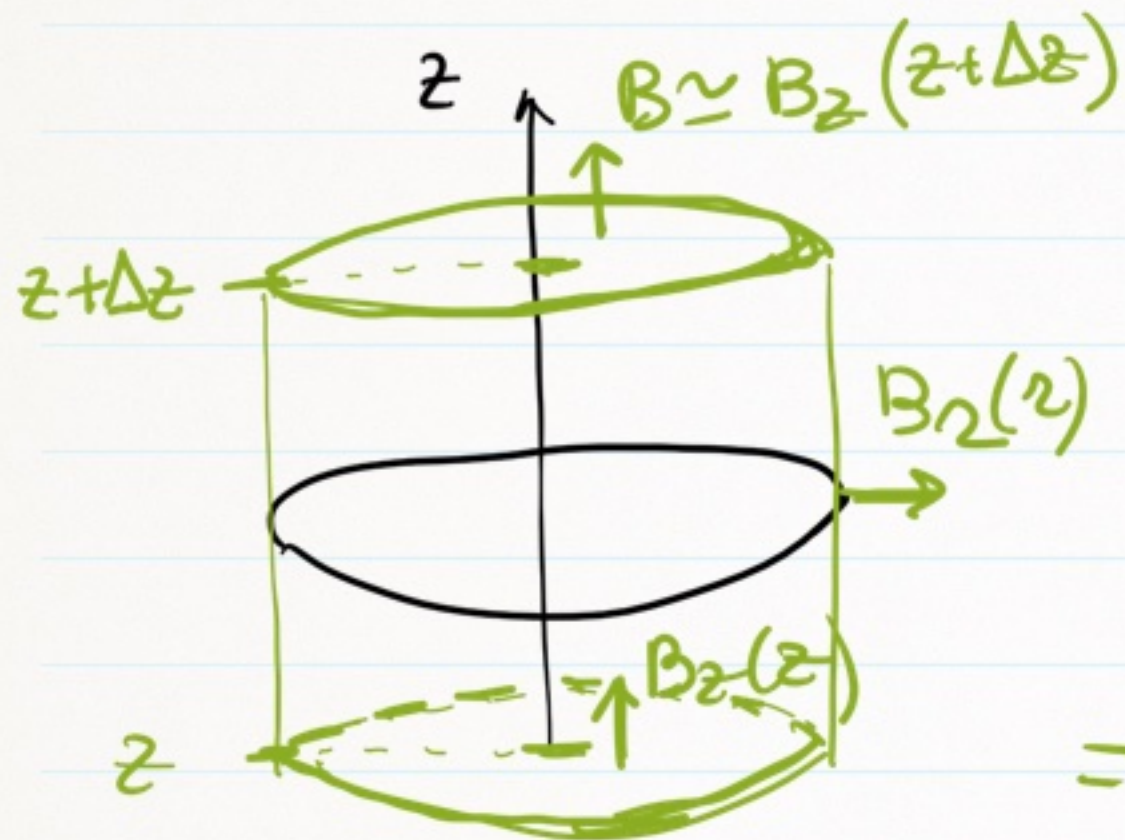
coefficiente di
attrito e.m.

(c) $\vec{F}_{ext} \nparallel \vec{v}$, e $\vec{F}$?

$$d\vec{F} = I_i \cdot d\vec{\ell} \times \vec{B}$$



$$\vec{F} = I_i \oint_{\text{spine}} d\vec{\ell} \times \vec{B} \longrightarrow \text{per risolvere mi serve } B_r$$



Legge di Gauss :

$$\vec{\nabla} \cdot \vec{B} = 0$$

$$\oint_{\text{cilindro}} \vec{B} \cdot d\vec{S} = 0$$

$$= 2\pi r \Delta z \cdot B_r(r) +$$

$$+ \pi r^2 [B_z(z + \Delta z) - B_z(z)]$$

$$\Rightarrow B_r(r) = - \pi r^2 \cdot \frac{B_z(z + \Delta z) - B_z(z)}{2\pi r \Delta z}$$

$$\approx \frac{\partial B_z}{\partial z} = \frac{B_0}{L}$$

$$B_r(r) = - \frac{r B_0}{2L}$$

$$\text{@ } r = a \longrightarrow \vec{B}_r = - \frac{B_0 a}{2L} \vec{u}_r$$

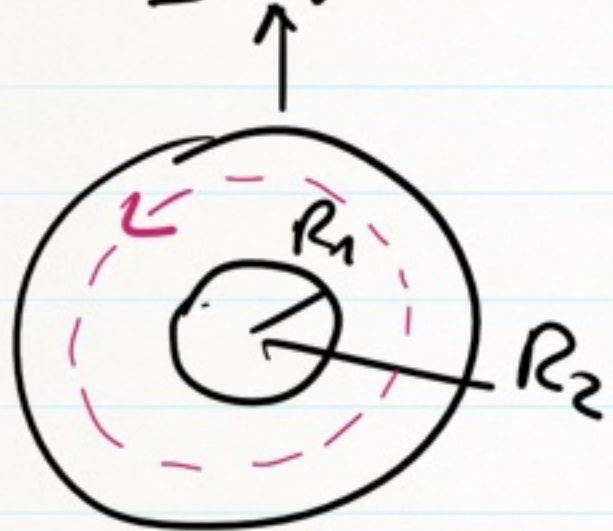
$$\vec{A} = I_i \oint_{\text{spine}} d\vec{l} \times \vec{B}_2(a) =$$

$$= -\vec{u}_z \frac{\pi a^2 B_0 v}{L R} \cdot \frac{B_0 a}{2L} \oint_{\text{spine}} dl$$

$$= -\vec{u}_z \left(\frac{\pi a^2 B_0}{L} \right)^2 \cdot \frac{v}{R}$$

Ex. 4

1 $\oint \vec{B} = L \cdot \vec{I}$



$0 < r < R_1$

$B = 0$

$R_1 < r < R_2$

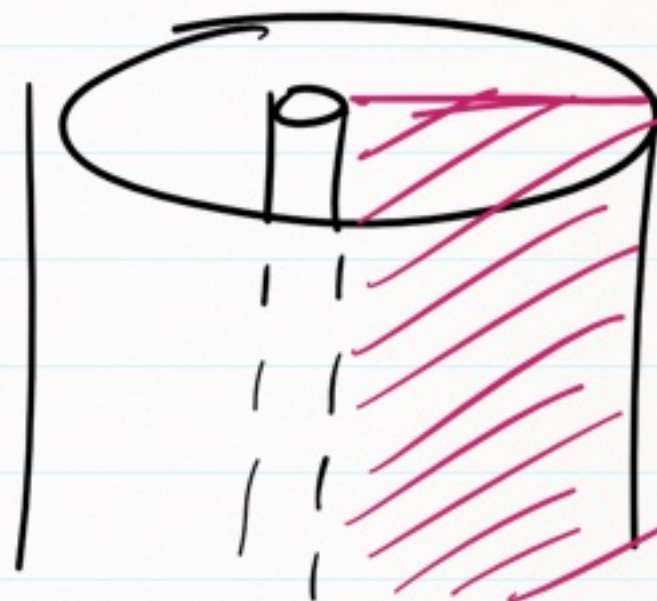
$B = \frac{\mu_0 I}{2\pi r}$

$r > R_2$

$B = 0$

$$\oint \vec{B} = \iint_S \vec{B} \cdot \vec{n} dS$$

$dS = h \cdot dz$



$$\Phi(\vec{B}) = \iint_S B \, dS = \frac{\mu_0 I h}{2\pi} \int_{R_1}^{R_2} \frac{dr}{r}$$

$$= \left(\frac{\mu_0 I h}{2\pi} \cdot \ln \frac{R_2}{R_1} \right) L$$

$$\underline{\underline{U_m = \frac{1}{2} L I^2}}$$

$$U_m = \iiint_V \frac{1}{2} \vec{B} \cdot \vec{H} \, d\tau$$

$$H = I / 2\pi r, \quad d\tau = r \, dr \, d\varphi \, dz$$

$$U_m = \frac{1}{2} \frac{\mu_0 I^2}{4\pi^2} \left(\int_{R_1}^{R_2} \frac{dr}{r} \right) \left(\int_0^{2\pi} d\varphi \right) \left(\int_0^h dz \right)$$

= ...

$$L = \frac{\mu_0}{2\pi} h \ln \frac{R_2}{R_1}$$

3. utilizzato pot. vettore $\vec{A}$ ($\vec{\nabla} \times \vec{A} = \vec{B}$)

$$\Phi(\vec{B}) = \iint_S \vec{B} \cdot \vec{n} \, dS = \int_{\gamma = \partial^+ S} \vec{A} \cdot d\vec{\ell}$$

$$\bullet \quad \vec{\nabla} \cdot \vec{B} = 0 \rightarrow \vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$$

$$\bullet \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{j} \rightarrow \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = -\nabla^2 \vec{A}$$

$$\bullet \nabla^2 \vec{A} = -\mu_0 \vec{j} \quad \vec{A}, \mu_0, (\vec{j})$$

$$\bullet \nabla^2 V = -\frac{\rho}{\epsilon_0} \quad V, \frac{1}{\epsilon_0}, (\rho)$$

potenziale
scalare (elettrico)

Costruisco $\vec{A}(r)$ in analogia con $V(r)$:

- per trovare $V(r)$ misuro $E(r)$
- per trovare $E(r)$ uso la Legge di Gauss

N.B.: $A(r) \equiv A_z(r)$

$$\oint (\vec{E}) = \frac{Q_{int}}{\epsilon_0}$$

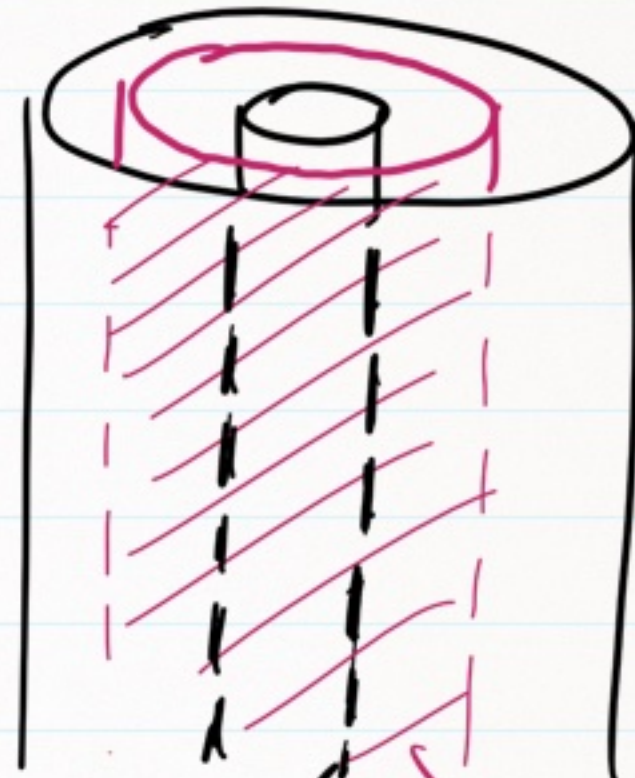
$$E(r) \cdot \underbrace{2\pi r h}_S = \frac{\lambda \cdot h}{\epsilon_0}$$

$$E(r) = \frac{\lambda}{2\pi\epsilon_0 r}, \quad R_1 < r < R_2$$

$$V(r) - V(R_2) = \int_r^{R_2} E(r') dr'$$

pongo $V(R_2) = \text{cost.} = 0$

$$V(r) = \frac{\lambda}{2\pi\epsilon_0} \cdot \ln\left(\frac{R_2}{r}\right)$$



$\lambda = \frac{Q_{int}}{h}$ *densità lineare di carica elettrica*
per $\oint(\vec{E})$

$$V = \frac{\lambda}{2\pi\epsilon_0} \cdot \ln\left(\frac{R_2}{r}\right)$$

costruisco $A_z(r)$
 per analogia

$$A_z(r) = \frac{\mu_0 I}{2\pi} \cdot \ln\left(\frac{R_2}{r}\right)$$

$$\cdot \frac{1}{\epsilon_0} \rightarrow \mu_0$$

$$\cdot \rho \rightarrow \vec{j} \quad ?$$

$$\underline{\lambda = \rho \Sigma} \rightarrow \underline{j \Sigma = I}$$

$$\oint (\vec{B}) = \oint_{\partial S} \vec{A} \cdot d\vec{\ell}$$

$$\oint (\vec{B}) = \int_{AB+BC+CD+DA} \vec{A} \cdot d\vec{\ell}$$

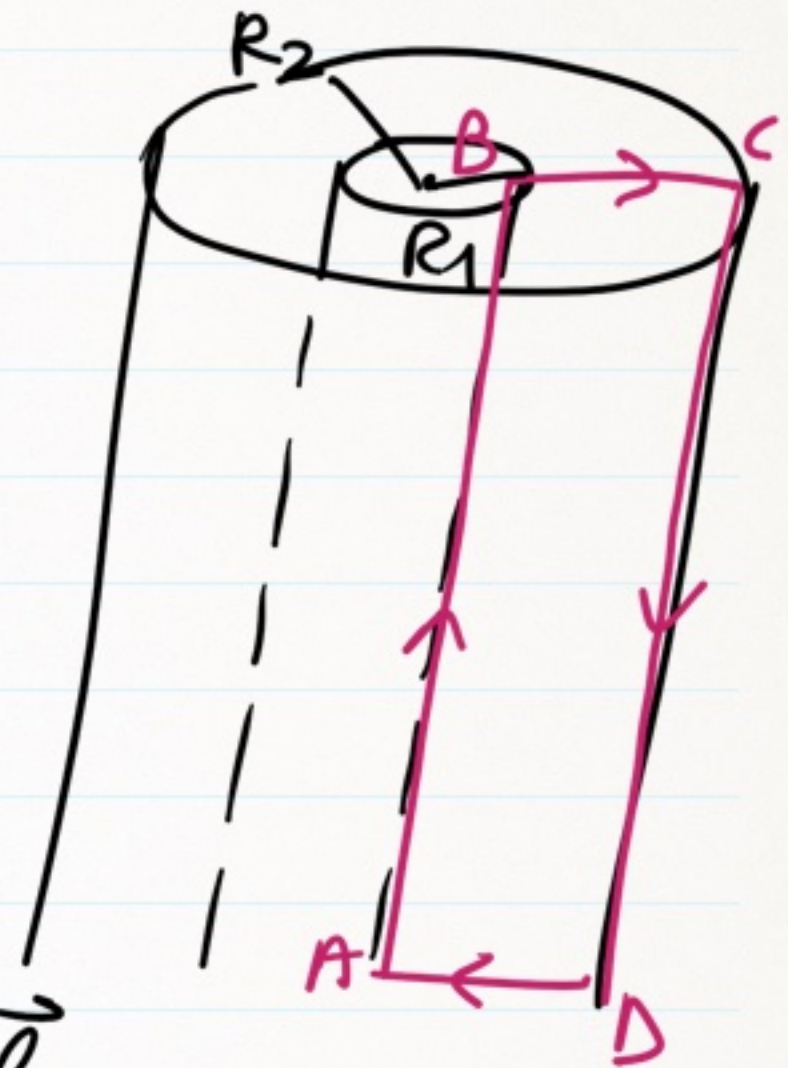
$$\int_{DA} \vec{A} \cdot d\vec{\ell} = - \int_{BC} \vec{A} \cdot d\vec{\ell}$$

quindi: $\oint (\vec{B}) = \int_{AB+CD} \vec{A} \cdot d\vec{\ell} =$

$$= A_z(R_1) \cdot h - A_z(R_2) \cdot h$$

sostituendo $A_z(z=R_1, R_2)$

$$= 0 \left(\dots \ln \frac{R_2}{R_1} \right)$$



$$\Phi(\vec{B}) = \frac{\mu_0 I h}{2\pi} \ln \frac{R_2}{R_1}$$

quindi

$$L = \frac{\mu_0 h}{2\pi} \ln \frac{R_2}{R_1}$$